

Load-Aware A* Path Planning for Quadruped Robots

Day 20 of 30 · Week 3: Comprehensive Testing · Member 1 — Figures for Paper

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1. Nature of Work Done

Today's work builds on the finalised Methodology (v2.0) submitted on Day 17. Three interconnected tasks were completed:

- 1. Explaining Load-Aware A* Advantages** — A structured comparison was developed highlighting where and why Load-Aware A* outperforms Standard A* in payload-carrying quadruped navigation, covering stability routing, cost structure, admissibility, and scalability.
- 2. Providing Theoretical Justification** — The mathematical underpinning of the stability-integrated cost function was reviewed. Admissibility and consistency proofs were verified to confirm the heuristic remains valid after introducing the stability penalty $S(n)$. The CoM displacement model and Support Polygon Margin were formally connected to the A* cost function.
- 3. Reviewing Result Interpretations** — Preliminary result patterns across the 20 test scenarios were analysed. Key metrics — path divergence, mean stability penalty, minimum stability margin, and compute overhead — were interpreted in light of algorithm design choices.

2. Load-Aware A* — Key Advantages

Stability-aware routing

Actively routes around cells with high CoM displacement, reducing tip-over risk throughout the path — not just at endpoints.

Continuous cost gradient

$S(n)$ is graded, not binary. The robot traverses mildly unstable cells when no better alternative exists, avoiding unnecessary detours.

Single-pass global planning

Stability is embedded in the global search objective. No post-processing or separate stability filter is required after path generation.

Admissibility preserved

$h(n)$ is unchanged (Manhattan distance). The stability penalty is added only to $g(n)$, so the heuristic stays admissible.

Higher $M_{\min}$ values

Load-Aware paths show consistently higher minimum stability margins, directly reducing the probability of mechanical failure.

Payload scalability

As payload mass increases (3–9 kg), Load-Aware A* adapts its routing automatically without parameter changes.

3.Theoretical Justification

Core Equations — Methodology v2.0

$$f(n) = g(n) + \alpha \cdot S(n) + h(n) \quad (\text{Eq. 10})$$

Load-Aware A* evaluation function. $\alpha = 1.0$ (stability weight). When $\alpha = 0$, reduces exactly to Standard A*.

$$g(n_{\text{next}}) = g(n_{\text{cur}}) + 1 + \alpha \cdot S(n_{\text{next}}) \quad (\text{Eq. 11})$$

Modified step cost — stability penalty accumulated at each cell traversal.

$$S(n) = \|x_{\text{CoM}}(n) - x_{\text{c}}(n)\|_2 \quad (\text{Eq. 5})$$

Stability Penalty: Euclidean distance from CoM to support polygon centroid. $S(n) = 0 \rightarrow$ CoM perfectly centred.

$$M(n) = \min_i d(x_{\text{CoM}}(n), \text{edge}_i) \quad (\text{Eq. 6})$$

Stability Margin: min perpendicular distance from CoM to any polygon edge. $M(n) > 0.05$ m required for stability.

$$x_{\text{CoM}} = (m_b \cdot x_b + m_p \cdot x_{\text{payload}}) / (m_b + m_p) \quad (\text{Eq. 4})$$

Weighted CoM of body ($m_b = 10$ kg) and payload ($m_p = 3\text{--}9$ kg). Payload shifts CoM away from geometric centroid.

4.Admissibility & Consistency — 5-Step Proof

- 1 . Heuristic unchanged: $h(n) = |r - r_g| + |c - c_g|$ (Manhattan, Eq. 8). Admissible for 4-connected grids: $h(n) \leq h^*(n)$ for all n , since each step closes at most 1 unit of Manhattan distance. Load-Aware A* does not modify $h(n)$, so admissibility is preserved.
- 2 . Penalty in $g(n)$ only: The stability term $\alpha \cdot S(n)$ is added exclusively to the accumulated path cost $g(n)$. Adding non-negative penalties to $g(n)$ does not affect admissibility — it makes the algorithm prefer paths with lower accumulated stability cost.
- 3 . Optimality guarantee: Load-Aware A* is optimal w.r.t. the combined cost $f = g + \alpha \cdot S + h$. It finds the path minimising total accumulated cost (length + stability penalty) — the intended design goal.
- 4 . When $\alpha = 0$: The algorithm reduces exactly to Standard A* (Eq. 10 collapses to $f = g + h$). This confirms backwards compatibility and validates Standard A* as the correct zero-payload baseline.
- 5 . S_{map} precomputation: All $S(n)$ values are computed in $O(R \times C) = O(100)$ before search begins (Eq. 12). Each lookup during A* is $O(1)$, meaning stability integration adds negligible per-node overhead.

5.Result Interpretations

Metric-by-Metric Interpretation (20 Scenarios)

D

Path Divergence: When $D = \text{True}$, Load-Aware A^* chose a different route from Standard A^* . Expected in scenarios with $m_p \geq 5$ kg where CoM is significantly displaced. Low divergence at $m_p = 3$ kg validates $\alpha = 1.0$ — the minimum weight producing consistent re-routing.

dL

Extra Steps ($L_{LA} - L_{std}$): Positive dL means Load-Aware A^* accepted a longer path to avoid high- $S(n)$ regions. Small dL (1–2 steps) with large improvement in S_{avg} or M_{min} represents an excellent trade-off. Large dL with minimal gain would question $\alpha = 1.0$.

S

Mean Stability Penalty (S_{avg}): Lower S_{avg} on Load-Aware paths directly confirms the algorithm routes through cells closer to the polygon centroid. This is the primary effectiveness metric for the algorithm design.

M

Minimum Margin (M_{min}): Safety-critical metric. If $M_{min} > 0.05$ m for Load-Aware A^* and $M_{min} \leq 0.05$ m for Standard A^* , the result strongly demonstrates practical advantage. Even small improvements in M_{min} carry significant mechanical safety implications.

T

Compute Overhead (dT%): $O(1)$ S_{map} lookup ensures dT% should be very small (<5%). Large dT% may indicate the open list grew due to modified costs causing more tie-breaking expansions — worth investigating per scenario.

N

Nodes Expanded: Load-Aware A^* may expand more or fewer nodes. In structured grids with clear stability gradients, early commitment to low-cost paths reduces N . In cluttered environments, stability penalties may create more cost ties, expanding N slightly.